Can Remote Sensing Replace the Map? Learning Multi-Modal Travel Speed from Overhead LiDAR, Imagery, and Crowd-Sourced GPS Traces

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1. Project Description and Goals

The problem. In 1993 Waldo Tobler proposed a hiking speed function that remains the default in GIS today [1]:

$$v = 6\exp(-3.5\,|s + 0.05|)\,, \tag{1}$$

with v in km/h and s the slope; other modes are handled by bolting on constant multipliers ($\times 3/5$ off-path, $\times 5/4$ on horseback). Nearly every travel-time and accessibility model in geography still runs on this formula or a close relative, yet it was fit to figures from a Swiss cartography textbook rather than to observed movement at scale. Two communities have circled the problem without meeting: GIS keeps Tobler because it drops into a cost-distance raster, while deep travel-time estimation works on road networks with dense traffic data and ignores terrain. The closest prior work, HikingTTE [3], learns hiking travel times with an LSTM and per-hiker personalization, but conditions on trajectories alone — no vegetation structure, no remote sensing.

What I propose to build. OpenStreetMap (OSM) sometimes records a trail's `surface`, `sac_scale`, and `width`. These tags predict speed well, and they are missing precisely where prediction is most valuable, so:

can remotely sensed terrain and vegetation structure substitute for trail attributes that a map does not have? If overhead LiDAR and imagery recover what OSM tags buy, travel speed becomes predictable anywhere sub-meter data exists, with no map at all — the input condition downstream off-road travel-time modeling in the lab requires. The experiment is a four-arm comparison. Every arm predicts segment speed and they differ only in what they may observe: *Tobler* sees slope alone with the fixed 1993 coefficients; *Tags* sees OSM trail attributes plus slope; *Remote sensing* sees LiDAR and imagery covariates plus slope but no OSM tags; *Both* sees everything. The contribution lives in whether the remote-sensing arm approaches the tags arm; a large residual gap is also worth publishing, since it would identify which trail properties matter for speed but are invisible from above.

Minimum goal (realistic). A cleaned, stop-filtered, map-matched trajectory dataset over one high-relief forested region, joined to 1 m terrain and vegetation covariates, plus the empirical slope–speed curve and an honest comparison against Eq. (1). This alone is a complete contribution, and it is what I commit to.

Maximum goal (ambitious). The full four-arm comparison with heteroscedastic heads predicting speed *and* variance; the directional asymmetry measured on paired out-and-back traversals; and per-mode curves testing whether Tobler's multiplicative treatment of mode holds at all. A four-page paper submitted to IGARSS 2027 (deadline 11 January 2027) and entered in the Student Paper Competition.

2. Relationship to Prior Work and My Background

Relationship to RasterNet [2]. The most directly relevant paper is RasterNet, from Prof. Jacobs' group, published at the EARTHVISION workshop at CVPR 2020. It estimates roadway free-flow speed by fusing overhead imagery with aerial LiDAR in a geospatially consistent raster, explicitly to avoid hand-measured geometric road features that are unavailable for most roads.

My project shares that premise — regress speed from overhead LiDAR and imagery when curated attributes are missing — and moves it where it binds harder. The domain is on-trail pedestrian and multi-modal travel, where slope is the dominant effect rather than a nuisance variable. Supervision comes from crowd-sourced OSM traces rather than commercial speed records, so dataset construction (stop detection, map-matching under canopy, mode inference) *is* the contribution rather than preprocessing. And I add a distributional output and a classical parametric baseline, since speed is only useful downstream if its uncertainty is calibrated. RasterNet's raster-alignment strategy is my reference design for the covariate stack.

What I already know. My prior research is in computer vision: Att-CapsViT, an attention-based CapsNet/ViT hybrid for handwritten Chinese character recognition (AIPMV 2024), and a comparative study of CNN versus pre-trained VGG16 for image classification (ICPPOE 2023). I also built a CNN classifier for galaxy morphology. I am comfortable with PyTorch, CNN and transformer architectures, transfer learning, ablation design, and Python/SQL/R tooling.

Where the line falls. None of that is this work. All

three prior projects were supervised classification on curated benchmarks — the dataset arrived ready. Here it does not exist, and building it is most of the semester. Geospatial data engineering is entirely new to me: OSM APIs, map-matching, LiDAR point clouds, coordinate reference systems, Earth Engine. So is probabilistic regression with proper scoring rules [6].

3. Resources

Data. GPS traces from the OSM endpoint `/api/0.6/trackpoints?bbox=` (capped at $0.25°$ per box, 5,000 points per page), so a region means 20–30 tiled requests, cached on first fetch. Trail graph from Geofabrik extracts via `osmium`/`pyrosm`/`geopandas`. Elevation from USGS 3DEP 1 m in Earth Engine, with point clouds from OpenTopography where vegetation structure is needed — a bare-earth DEM has no understory, and understory density is what makes travel slow [4]. Imagery from NAIP (1 m, RGB+NIR), canopy from NLCD, plus point-sampled AlphaEarth embeddings.

Region. Candidates for the weeks 1–2 audit: White Mountains (NH), Cascades (WA), Colorado Front Range. The criteria are firm even if the choice is not — real relief, a dense multi-use trail network, forest cover, verified 3DEP 1 m coverage, and enough ordered traces.

Implementation and outside code. Python and PyTorch; `leuvenmapmatching` or an HMM matcher following Newson and Krumm [5]; `gdaldem` and the Relief Visualization Toolbox for terrain derivatives; PDAL/`laspy` for point clouds. Gradient-boosted trees are the serious tabular baseline, and at this sample size may be harder to beat than a neural network.

Compute. Modest — 10–30 tabular covariates and a scalar target, with effective sample size set by the number of *traces*, not segment-rows. The real cost is I/O and storage, and point-cloud processing is the slowest step, contained by extracting vegetation metrics only within a buffer around traced trails.

Access I do not yet have. Flagging these now, since they are the likeliest cause of a stall: (i) a Google Earth Engine account, which requires registration and approval; (ii) lab compute and storage credentials; (iii) lab swipe access, pending my CSE 5999 enrollment paperwork. None blocks the week 1–3 trace reconnaissance, which runs against the public OSM API from my own machine, but all three are needed before covariate extraction in week 7.

4. Reservations

The hardest part of this project is not the modeling; it is trace handling, and I expect it to take six weeks.

The gating risk. OSM exposes traces at four visibility levels, and only `identifiable` and `trackable` preserve temporal ordering; `public` traces return points sorted by coordinate, destroying the ordering speed is computed from. Nobody has published what fraction in a region is usable. If the audit comes back mostly scrambled, the right response is to change region immediately — hence a hard gate in week 3, not a discovery in week 8.

Technical hazards. Map-matching is harder on trails than roads: the graph is sparse, trails converge near junctions, and canopy degrades GPS accuracy right where the trails are. Stop detection is a single thresholding choice that will move my headline numbers more than any model change, so it needs an explicit rule consistently applied with the dropped fraction reported. Mode inference is circular if done naively, so I will fit mode-conditional distributions only on exclusive-permission segments, then classify ambiguous ones. Evaluation must split by trace and trail system, never point.

One hazard I cannot fix: people who upload traces to OSM are mappers and enthusiasts, not a random sample of hikers, so the model is a prior over that population and the limitations will say so plainly.

Why the minimum goal survives. The stretch items are individually cheap and collectively how a semester overruns, so the plan slips backward, never forward: if nothing after week 11 lands, the dataset plus the Tobler comparison is still a paper.

5. Mentor and Meeting Plan

I am mentored by **Danni Beaulieu** and **Tianyi Gao**, both PhD students in Prof. Nathan Jacobs' lab (CSE, McKelvey School of Engineering); Prof. Jacobs is my faculty advisor. We first met in person on 24 August 2026 and have corresponded since by email and Slack, where Danni scoped the semester plan this project follows.

Going forward we will meet **weekly**. I will bring what I did, what broke, and one plot each time — a plot that looks wrong is more useful than no plot. I will escalate rather than grind if stuck on the same problem for more than two days, and will keep a decisions log of every threshold, filter, and split boundary for the data card and methods section.

References

[1] W. Tobler. Three presentations on geographical analysis and modeling. Tech. Rep. 93-1, NCGIA, 1993.

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[3] M. Asako et al. Deep learning-based travel time estimation in hiking with consideration of individual walking ability. *Cybernetics and Information Technologies*, 2024.

[4] M. J. Campbell et al. Quantifying understory vegetation density using small-footprint airborne lidar. *Remote Sens. Environ.*, 215:330–342, 2018.

[5] P. Newson and J. Krumm. Hidden Markov map matching through noise and sparseness. In *ACM SIGSPATIAL*, 2009.

[6] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *JASA*, 102(477), 2007.